

Challenge (Habanero)

- Q1.** What is the x -coordinate of the solution to the system:
 $2x + 3y = 7$ and $x - 2y = -3$?
(A) -1
(B) 1
(C) 2
(D) 3
(E) 4
- Q2.** A gamer's score S is determined by $S = 2x + 5y$, where x is the number of levels completed and y is the number of coins collected. If $S = 17$, and $x - 3y = -2$, what is y ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q3.** In a coding challenge, the number of points P earned is given by $P = 3x - 2y$, where x is the number of tasks completed and y is the number of errors made. If $P = 12$, and $2x + y = 10$, what is x ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q4.** The data usage D of a gaming app is given by $D = x + 2y$, where x is the number of hours played and y is the number of players online. If $D = 12$, and $x - y = 2$, what is x ?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q5.** What is the y -coordinate of the solution to the system:
 $x + 4y = 11$ and $3x - 2y = 5$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q6.** A scoreboard displays the number of wins W and losses L of a team, where $W = 2x + y$ and $L = x - 2y$. If $W = 10$ and $L = 2$, what is y ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q7.** The cost C of developing a game is given by $C = 3x + 2y$, where x is the number of developers and y is the number of months. If $C = 24$, and $2x - y = 4$, what is y ?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q8.** A player's ranking R is determined by $R = x - 3y$, where x is the number of matches won and y is the number of matches lost. If $R = 5$, and $x + 2y = 12$, what is y ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations: $x + 2y = 7$ and $3x - y = 5$. Show all work and explain your solution.
- Q10.** A game developer wants to create a new level with a specific number of coins and power-ups. The number of coins C is given by $C = 2x + y$, and the number of power-ups P is given by $P = x - 2y$. If the developer wants $C = 15$ and $P = 3$, find the values of x and y . Show all work and explain your solution.

Chef's Tips

- Tip 1: Encourage students to graph the systems of equations to visualize the solutions.
- Tip 2: Remind students to check their work by plugging the solutions back into the original equations.
- Tip 3: Emphasize the importance of labeling the axes and including a title when graphing the systems.